Assessment Tool- 3

Short Answer Test

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5) Learing Problems and Perspectives.

ML mainly has three types of learning:-

1) Supervised Learning:- The model learn from labelled data containing input and expected output.

2) Unsupervised Learning:- The model learns patterns from data without labelled outputs.

Ex:- clustering & dimensionality reduction.

3) Reinforcement learning:- An agent learns by interacting with an environment and receving rewards or penalties.

Ex:- Game playing and robotics.

Prespectives of learning:-

=> symbolic approach:- Uses rules, logic, and knowledge representation.

=> statistical approach:- Uses probability and mathematical model to learn patterns from data.

key issues:- Poor data quality, overfitting, insufficient training data, & high computational complexity can affect model performance.

Conclusion:- choosing the appropriate learning type and mapping data quality are important for building an effective ML system.

9) Decision Tree Learning :- A decision tree is a supervised learning algorithm used mainly for classification regression.

→ It consists of root, internal nodes, branches and leaf nodes.
→ Each internal node represents a feature (or) condition
→ Each branch represents a decision.
→ Leaf nodes represent the final prediction.

Entropy measures the impurity (or) uncertainty in the data

$$Entropy(S) = -\sum p_i \log_2 p_i$$

Information Gain :- measures how much uncertainty is reduced after splitting the data:

IG = Entropy (parent) − weighted Entropy (children)

The feature with the highest information gain is generally selected for splitting.

Pruning removes unnecessary branches to reduce overfitting.

Advantages :- Easy to understand, interpretable and requires little data preprocessing.

Limitations :- Can overfit and may become complex for large datasets.

11) Artificial Neural Network :- It is a ML model inspired by the human brain.

It mainly consists of.

1) Input layer :- receives input features.
2) Hidden layers(s) - processes the input
3) Output layer - produces the final prediction.

Each neuron calculates a weighted sum of inputs and applies an activation function such as ReLU

sigmoid or
Backpro
it

sigmoid or tanh.
Backpropagation calculates the error and propagates it backward through the network to update weights.
Gradient Descent minimizes the error by adjusting weights in the direction that reduces the loss.

Advantages of deep learning:-
* Automatically learns useful features.
* Handles large and complex datasets.
* Provides high performance for images, speech & text.

Application:- Image recognition, speech recognition, medical diagnosis, autonomous vehicales, and natural language processing.

(2) Support vector machine:- It is a supervised learning algorithm mainly used for classification.
The main objective is to find the best hyperplane that separates different classes.
* Hyperplane:- Decision boundary separating classes.
* Margin:- Distance between the hyperplane & the nearest data points.
* Support vectors:- Data points closest to the hyperplane.
SVM tries to maximize the margin, which generally improves classification performance.
For non-linear data, SVM uses kernel function to transform data into a higher-dimensional space.
Common kernels include:-
* Linear kernel
* polynomial kernel

* Radial Basis function (RBF)
* Sigmoid kernel

Advantages:- Effective in high-dimensional data & works well with clear class boundaries

Applications:- Image classification, text classification, face recognition, and pattern recognition.

(15) Model Evaluation Techniques:-

Model evaluation checks how well a machine learning modern performs.

→ Train-Test split:- Divides data into training & testing data.

→ Cross-Validation:- Divides data into multiple parts & train & tests repeatedly.

→ Bootstrapping:- Creates multiple samples from the dataset for evaluation.

Classification metrics:- Accuracy, Precision, Recall, F1-score.

Regression metrics:- MAE, MSE, RMSE & R^2.

Proper validation helps avoid overfitting and ensures that the model performs well on unseen data.